

$\mathcal{ALCI}_{\text{RCC5}}$: Audit of the Decidability Claim and Two Partial Results Beyond the $\forall\text{PO}$ -Free Fragment

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Abstract

This report examines the current project snapshot supplied as `alcircc5-master(1).zip`, using its main Markdown pages and their current paper and Lean references. No gap was found in the present $\forall\text{PO}$ -free concept-satisfiability argument or in the inspected dependencies of its Lean capstones. This was a source audit, not a fresh Lean kernel run. Unrestricted $\mathcal{ALCI}_{\text{RCC5}}$ remains unresolved.

Two mathematical results are proved. First, for a fixed strict part order, disjointness has a unique largest admissible interpretation: two points are disjoint exactly when they have no common lower bound. Principal down-sets realize it, and the transformation preserves every NNF concept without $\exists\text{PO}$ or $\forall\text{DR}$. Second, for NNF concepts without $\exists\text{PP}$, $\exists\text{PO}$, or $\forall\text{DR}$, replacing each $\forall\text{PO}$ restriction by $\top$ preserves satisfiability. A fresh downward witness forest proves the second result and yields a decidable syntactic extension of the certified fragment. Explicit counterexamples delimit both constructions. These new proofs have not been formalized in Lean and do not decide the unrestricted logic. The report concludes with a separately identified, subjective outlook on the prospect of an AI-assisted resolution over the next two to five years.

1 Scope and conclusions

The authoritative status documents for this review are the supplied `README.md`, the approach pages for the cone scheme and Lean/F6, and the current sources `papers/pofree_fragment_arxiv.tex` and `papers/overview_arxiv.tex`. The relevant formal development is `formal/POFreeLift.lean`. Historical failures were used only to identify a potential source of error; they were not treated as current proofs.

The subject is *single-concept satisfiability* under the abstract RCC5 composition-table semantics with strong equality. The roles are the five atomic RCC5 relations. The report does not assert a decision result for general TBoxes or for a fixed domain containing *all* regions of a particular space. All syntactic restrictions below are imposed after conversion to negation normal form (NNF).

Claim	Assessment in this review
$\forall$ PO-free satisfiability	No gap found in the current proof or inspected Lean dependencies. No fresh kernel build was performed.
Full unrestricted satisfiability	Still open. The certificate-completeness premise required by the full-language reduction is unproved.
Principal-down-set normalization	Proved here for the NNF fragment without $\exists$ PO and $\forall$ DR. No decidability conclusion for that whole fragment.
PO erasure	Proved here when $\exists$ PP, $\exists$ PO, and $\forall$ DR are absent. Gives a decidable syntactic extension.

The two new results are relative to the established RCC5 semantics. They are mathematical results of this review, not claims of literature priority. They have been independently checked for the relation orientations and boundary cases, but have not been added to the Lean artifact.

2 Semantics and the order presentation

Let $\mathcal{R} = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}$. An RCC5 frame has a nonempty carrier W and a total relation labelling $\rho : W^2 \rightarrow \mathcal{R}$, with identity as EQ, the usual converse operation, and the RCC5 composition condition. Here $x \text{ PP } y$ means that x is a *proper part of* y ; thus a PP witness is a superpart and a PPI witness is a part.

An interpretation adds a valuation $V(A) \subseteq W$ for each concept name. NNF concepts have grammar

$$C ::= \top \mid \perp \mid A \mid \neg A \mid C \sqcap C \mid C \sqcup C \mid \exists R.C \mid \forall R.C, \quad R \in \mathcal{R}.$$

The modal clauses are

$$\begin{aligned} \mathcal{M}, x \models \exists R.D &\iff \exists y \in W [\rho(x, y) = R \wedge \mathcal{M}, y \models D], \\ \mathcal{M}, x \models \forall R.D &\iff \forall y \in W [\rho(x, y) = R \implies \mathcal{M}, y \models D]. \end{aligned}$$

Satisfiability means truth at a point of some such interpretation.

Definition 2.1 (Ordered-disjoint structure). An ordered-disjoint structure is $(W, <, \#)$ where $<$ is a strict partial order and $\#$ is symmetric and irreflexive, with the following properties. Writing $\leq$ for the reflexive closure of $<$,

$$x < y \implies \neg(x \# y), \tag{1}$$

$$x \# y, \quad u \leq x, \quad v \leq y \implies u \# v. \tag{2}$$

Its induced labelling gives EQ to equality, PP to $<$, PPI to $>$, DR to $\#$, and PO to the remaining pairs.

Theorem 2.2 (Established RCC5 normal form). *The induced labelling of every ordered-disjoint structure is an RCC5 frame. Conversely, reading $<$ as PP and $\#$ as DR in any strong-EQ RCC5 frame yields an ordered-disjoint structure with the original labelling. The abstract frames and families of nonempty, pairwise distinct sets have the same concept satisfiability.*

This theorem is an established input, proved in the current project [2, 4]. Transitivity comes from $\text{PP} \circ \text{PP} = \{\text{PP}\}$; downward inheritance follows from $\text{PP} \circ \text{DR} = \{\text{DR}\}$ and $\text{DR} \circ \text{PPI} = \{\text{DR}\}$. The

converse is the checked composition-table case analysis. The project's general set representation retains overlap even when no common part is represented in the carrier; this distinction will matter below.

Under set semantics, the domain is a chosen family of nonempty sets. Equality, strict inclusion, converse inclusion and disjointness interpret the first four applicable cases; PO means nonempty intersection and incomparability under inclusion. A shared point of the ambient set universe need not itself occur as a named region in the domain.

3 Audit of the existing decidability argument

3.1 Finite signatures and unconditional completeness

For a fixed NNF concept C_0 , the current cone procedure takes support types $T \subseteq \text{cl}(C_0)$ and signatures $q = (T, S)$, where S is a set of candidate types allowed strictly below q . Let $\text{cone}(q) = \{T\} \cup S$ and $\text{Bod}_R(T) = \{D : \forall R.D \in T\}$. Static admissibility checks the local Boolean and EQ conditions, and for each $U \in S$ requires

$$\text{Bod}_{\text{PP}}(U) \subseteq T, \quad \text{Bod}_{\text{PPI}}(T) \subseteq U.$$

A target $q' = (T', S')$ for a non-EQ demand $\exists R.D \in T$ must contain D in T' and satisfy the appropriate compatibility condition:

R	Compatibility condition
PP	$\text{cone}(q) \subseteq S'$.
PPI	$\text{cone}(q') \subseteq S$.
DR	For all $U \in \text{cone}(q)$ and $U' \in \text{cone}(q')$, $\text{Bod}_{\text{DR}}(U) \subseteq U'$ and $\text{Bod}_{\text{DR}}(U') \subseteq U$.
PO	$\text{Bod}_{\text{PO}}(T) \subseteq T'$ and $\text{Bod}_{\text{PO}}(T') \subseteq T$.

Pruning removes signatures that lack a compatible target for some demand. The operator is monotone and reductive on a finite initial signature set.

In any actual model, take the complete closure type of a point and the set of closure types realized at its proper parts. These signatures are statically admissible and support one another using the model's witnesses. They consequently survive every pruning round. This direction does *not* assume $\forall\text{PO}$ -freeness. The checked declaration is `coneScheme_complete`.

3.2 Fresh occurrences and fragment soundness

The converse unfolds the surviving finite control states to fresh occurrences. Part births generate a strict order; DR births generate disjointness together with all its downward consequences. Residual pairs are PO. The height and vertical-component invariants establish a genuine ordered-disjoint structure. Cone propagation handles all PP and PPI universals, and the cross-cone test at every DR birth handles all DR universals. Witness children serve existential demands.

The truth lemma uses the fragment hypothesis precisely when it excludes $\forall\text{PO}.D$ from occurrence labels. Most PO pairs were not created as witness edges: they are residual pairs between branches. A universal over PO would constrain these pairs too. This step appears in `unf_truth`, and gives `coneScheme_sound`. The finite fixed point then yields `decidableSat_cone`.

The audit found no missing premise in this current argument. In particular, the DR test compares entire downward cones, and the unfolding does not equate two occurrences simply because their signatures agree. Both details are necessary. The source also proves normalization from raw negation in both polarities. The raw fragment excludes positive $\forall\text{PO}$ and negative $\exists\text{PO}$; merely checking the spelling of modalities before NNF would be incorrect.

3.3 Verification strength and limits

The audit inspected the capstone statements and their proof dependencies, the semantics, the raw polarity test, and the abstract-to-set equivalence. No executable `sorry`, `admit`, or new user axiom was identified in the current formal files. A separate finite enumeration recovered all 25 displayed RCC5 composition cells from nonempty subsets of a four-element universe.

Lean was unavailable in the review environment, and the archive contained no compiled `.olean` artifacts. Accordingly, this report does not claim an independent kernel recompilation or fresh axiom output. The repository reports such certification; the present review found no source-level defect in the relevant argument. The theorem `satisfiable_iff_set` concerns arbitrary set families. The paper’s further regular-closed-region corollary uses the representation theorems of Lutz and Wolter [5] externally.

4 Maximal disjointness and principal down-sets

Fix a nonempty strict partial order $(W, <)$. Write $x \parallel y$ when x, y are incomparable under its reflexive closure. Define

$$x \#_{\max} y \iff \neg \exists z \in W (z \leq x \wedge z \leq y). \quad (3)$$

Lemma 4.1 (Largest admissible disjointness). *The relation $\#_{\max}$ is the unique largest relation, under inclusion, which makes $(W, <, \#)$ an ordered-disjoint structure.*

Proof. Symmetry is immediate. Irreflexivity holds because x is a lower bound of x, x . If $x \leq y$, then x is a common lower bound, so comparable pairs are not in $\#_{\max}$. For downward inheritance, suppose $x \#_{\max} y$, $u \leq x$ and $v \leq y$. A common lower bound of u, v would, by transitivity, be a common lower bound of x, y , a contradiction. Thus $\#_{\max}$ is admissible.

If $\#$ is any admissible relation and $x \# y$, a common lower bound $z \leq x, y$ would yield $z \# z$ by downward inheritance, contradicting irreflexivity. Hence $\# \subseteq \#_{\max}$, proving maximality and uniqueness of the largest relation. $\square$

For each $x \in W$, put $P_x = \{z \in W : z \leq x\}$.

Lemma 4.2 (Exact set representation). *The family $(P_x)_{x \in W}$ consists of nonempty, pairwise distinct sets, and*

$$P_x \subseteq P_y \iff x \leq y, \quad P_x \cap P_y = \emptyset \iff x \#_{\max} y.$$

Consequently its RCC5 labelling preserves the order and gives

$$\text{PO}_{\min}(x, y) \iff x \parallel y \wedge \exists z \in W (z \leq x \wedge z \leq y). \quad (4)$$

Proof. The element x belongs to P_x . If $x \leq y$, transitivity gives $P_x \subseteq P_y$; conversely, that inclusion puts x in P_y . Equality of the two sets therefore implies $x \leq y$ and $y \leq x$, hence $x = y$. Their intersection consists exactly of the common lower bounds. The RCC5 set clauses now give (4). $\square$

For an RCC5 interpretation $\mathcal{M}$, let $\mathcal{M}_{\max}$ be its principal-down-set interpretation, identified with the same index set W , and with the same unary valuations. Its order is unchanged and its disjointness is $\#_{\max}$.

Theorem 4.3 (Truth preservation under maximal disjointness). *Let C be an NNF concept with no occurrence of $\exists\text{PO}$ or $\forall\text{DR}$, including inside modal bodies. For every RCC5 interpretation $\mathcal{M}$ and $x \in W$,*

$$\mathcal{M}, x \models C \implies \mathcal{M}_{\max}, x \models C.$$

Therefore satisfiability for this fragment can be restricted to principal-down-set interpretations.

Proof. Lemmas 4.1–4.2 show that EQ, PP and PPI are unchanged, old DR edges remain DR, and only old PO edges can change, each either remaining PO or becoming DR. Induct on C . Literals, constants and Boolean connectives follow from the unchanged valuation and the induction hypothesis. For EQ, PP and PPI modalities, use their unchanged successor sets and the induction hypothesis on the body. An old DR existential witness is still a DR witness and still satisfies its body. Every new PO successor was an old PO successor, so a PO universal continues to hold by the induction hypothesis. The two modal cases that could fail are precisely the two excluded constructors. The normalized interpretation is itself an RCC5 interpretation, giving the converse implication for *existence* of a satisfying model without any additional argument. $\square$

Remark 4.4 (Normalization is not yet a decision procedure). The order can still force PO. For example,

$$\exists\text{PP}.(A \sqcap \forall\text{PP}.A) \sqcap \exists\text{PP}.(¬A \sqcap \forall\text{PP}.¬A)$$

forces distinct incomparable superparts a, b of the root: either order between them violates one of the PP universals. Their shared proper part excludes DR, so they must be PO. The normal form retains such overlaps.

5 A decidable fragment by erasing PO universals

Definition 5.1 (The downward-witness fragment). Let $\mathcal{D}$ be the class of NNF concepts with no occurrences of $\exists\text{PP}$, $\exists\text{PO}$ or $\forall\text{DR}$. All other constructors are permitted, including nested $\forall\text{PO}$. Define *erase* recursively by

$$\text{erase}(\forall\text{PO}.D) = \top,$$

and homomorphically on every other constructor. In particular, an erased PO restriction loses its entire body. Then $\text{erase}(C)$ is $\forall\text{PO}$ -free and $|\text{erase}(C)| \leq |C|$.

Lemma 5.2 (PO-empty witness-forest property). *Every satisfiable NNF concept E without $\forall\text{PO}$, $\exists\text{PP}$, $\exists\text{PO}$, or $\forall\text{DR}$ has a countable model in which PO is empty.*

Proof. Let $\mathcal{M}, x_0 \models E$. We construct occurrences in finitely many stages per depth. Each occurrence u carries a source point $f(u) \in W^{\mathcal{M}}$. Begin with u_0 and $f(u_0) = x_0$. For each already generated occurrence u and every existential subformula occurrence $e = \exists R.D$ of E with $\mathcal{M}, f(u) \models e$ and

$R \neq \text{EQ}$, choose a witness y in $\mathcal{M}$ and generate a fresh occurrence $u \cdot e$ with $f(u \cdot e) = y$. There are finitely many such requirements per occurrence, so the set U of all generated finite histories is countable. The only non-EQ roles needing witnesses are PPI and DR.

Retain the parent edge between u and $u \cdot e$ only when e has role PPI, and put $u \cdot e < u$. A DR child starts a new component with no vertical edge to its generating occurrence. The transitive closure of these retained edges is a strict order. Its components are downward rooted trees: every non-root has exactly one immediate superpart, and every point has a finite chain of ancestors to its component root.

Set $u \# v$ exactly when u, v are incomparable. This is downward hereditary. Indeed, if u', v' lie below incomparable u, v but are comparable, one of u', v' lies below both u, v . Its ancestors form a chain, forcing u, v to be comparable, a contradiction. Thus the order and $\#$ form an ordered-disjoint structure, with no residual PO pairs. Equivalently, the principal down-sets of its nodes are nested along branches and disjoint between incomparable nodes, directly supplying a set representation. Let $\mathcal{N}$ be this frame with $u \in V^{\mathcal{N}}(A)$ iff $f(u) \in V^{\mathcal{M}}(A)$.

Whenever $u < v$ in $\mathcal{N}$, the source points satisfy $f(u) \text{ PP } f(v)$ in $\mathcal{M}$: each edge has this property by the selected PPI witness, and the property composes transitively. Hence the map f preserves every new PP and PPI edge in its original direction. It also preserves equality. A generated DR child lies in a fresh component and therefore is DR from its generating point.

We now prove simultaneously, for every subformula D of E and every occurrence u , the implication

$$\mathcal{M}, f(u) \models D \implies \mathcal{N}, u \models D. \quad (5)$$

Induct on D . Literals and Boolean cases follow from the pulled-back valuation and the induction hypothesis. Non-EQ existential cases are served by their generated children, whose sources satisfy the body. EQ modalities reduce to the body at the same point. For a PP or PPI universal, each new successor has the same relation between its source points, so the old universal and the induction hypothesis apply. No other universal or existential case remains under the stated restrictions. Applying (5) to E at u_0 proves the result. $\square$

Theorem 5.3 (PO-erasure equivalence). *For every $C \in \mathcal{D}$,*

$$\text{Sat}(C) \iff \text{Sat}(\text{erase}(C)).$$

Proof. In NNF all compound constructors are monotone in their concept arguments. Replacing an entire $\forall\text{PO}$ subformula by $\top$ therefore preserves satisfaction, proving the forward implication.

Conversely, $\text{erase}(C)$ satisfies the hypotheses of Lemma 5.2. If it is satisfiable, it has a model $\mathcal{N}$ with empty PO. In any such model, for every concept D and every point u ,

$$\mathcal{N}, u \models D \iff \mathcal{N}, u \models \text{erase}(D).$$

This follows by structural induction: a PO universal and its replacement $\top$ are both true, and all remaining constructors commute with erasure. In particular, the point satisfying $\text{erase}(C)$ satisfies C . $\square$

Corollary 5.4 (Effective extension of the existing decision result). *Satisfiability for $\mathcal{D}$ is decidable. So is satisfiability for the union of $\mathcal{D}$ with the published $\forall\text{PO}$ -free fragment. This union is a strict syntactic extension of that fragment.*

Proof. Both syntactic membership tests are effective. For $C \in \mathcal{D}$, compute $\text{erase}(C)$ and apply the established $\forall\text{PO}$ -free decision procedure. Use the established procedure directly on the original fragment. The formula $\forall\text{PO}.A$ belongs to $\mathcal{D}$ but is not $\forall\text{PO}$ -free. $\square$

This extension does not add an irreducible PO-universal satisfiability constraint: those restrictions are erasable within $\mathcal{D}$. It is nevertheless a decision theorem about inputs outside the original syntactic fragment. Its new model-construction proof is not yet Lean checked. The same doubly exponential upper bound stated for the literal cone procedure applies after erasure, since erasure does not increase size; no new lower bound is asserted.

6 Counterexamples delimiting the constructions

Each excluded constructor in Theorem 5.3 is necessary for that unrestricted erasure rule. In the following examples, erasure is satisfiable while the original concept is unsatisfiable.

6.1 Permitting an existential PO restriction

The immediate example is

$$C_{\exists\text{PO}} = \exists\text{PO}.\top \sqcap \forall\text{PO}.\perp.$$

It is unsatisfiable, whereas its erasure $\exists\text{PO}.\top$ is satisfied by two overlapping, incomparable sets.

There is also a sharper boundary for Theorem 4.3:

$$C_{\text{optional}} = \forall\text{PPI}.A \sqcap \exists\text{PO}.\neg A. \quad (6)$$

The two-region domain $x = \{0, 1\}$, $y = \{1, 2\}$ satisfies this at x ; neither region has a represented proper part. In any satisfying model, a common lower bound of x and its PO witness y would be a strict part of both, forcing both A and $\neg A$. Thus the overlap must lack a represented common part. Such a pair is DR under principal down-sets. Consequently (6) has no principal-down-set model at all.

6.2 Permitting an existential PP restriction

Consider

$$\begin{aligned} C_{\exists\text{PP}} = & \exists\text{PP}.(A \sqcap \forall\text{PP}.A \sqcap \forall\text{PO}.A) \\ & \sqcap \exists\text{PP}.\neg A \sqcap \forall\text{PP}.\neg A. \end{aligned} \quad (7)$$

At a putative root x , choose the witnesses a, b with $A(a)$ and $\neg A(b)$. They are distinct. If $a \text{PP} b$, the universal at a forces $A(b)$; if $b \text{PP} a$, the universal at b forces $\neg A(a)$. They cannot be DR because their shared proper part x would then be disjoint from itself. Thus $a \text{PO} b$, but the PO universal at a again forces $A(b)$, a contradiction.

Its erasure has the three-region model

$$x = \{0\}, \quad a = \{0, 1\}, \quad b = \{0, 2\}, \quad V(A) = \{a\}.$$

The two PP universals are vacuous at the maximal regions a, b . This shows why upward witnesses can prevent a PO-empty forest construction.

6.3 Permitting a universal DR restriction

Consider

$$C_{\forall DR} = \exists PPI.(A \sqcap \forall PPI.A \sqcap \forall DR.A \sqcap \forall PO.A) \\ \sqcap \exists PPI.(\neg A \sqcap \forall PPI.\neg A). \quad (8)$$

Its two proper-part witnesses a, b are distinct. If $a < b$, the PPI universal at b contradicts $A(a)$; if $b < a$, the PPI universal at a contradicts $\neg A(b)$. A DR relation contradicts the DR universal at a , and a PO relation contradicts its PO universal. Every RCC5 value is excluded, so the concept is unsatisfiable.

The erasure has the model

$$x = \{0, 1, 2\}, \quad a = \{0, 1\}, \quad b = \{1, 2\}, \quad V(A) = \{a\}.$$

All displayed PPI and DR universals at a, b are vacuous. Moreover, *every* model of this erased concept forces a, b to be PO: the preceding case split still excludes EQ, PP, PPI and DR. A common lower bound would satisfy both A and $\neg A$ by their PPI universals. Thus the erased concept also has no principal-down-set model. This proves that admitting arbitrary $\forall DR$ breaks the normalization theorem even if $\exists PO$ remains absent.

7 Consequences for the unrestricted problem

The new normalization distinguishes two sources of overlap. For fixed order, an incomparable pair with a represented common lower bound is necessarily PO; this overlap survives every admissible choice of DR. Other PO pairs are optional relative to the order, although concepts can require them, as (6) and the erased form of (8) demonstrate. A complete construction for the unrestricted language must accommodate both kinds and the universal constraints on all resulting pairs.

The current cone scheme has no such complete joint control. Its full-language completeness direction provides a sound UNSAT test, but acceptance is not sufficient. The current focused paper records

$$C_{\text{bad}} = \exists PPI.(\exists PP.A) \sqcap \forall PP.\neg A \sqcap \forall PPI.\neg A \sqcap \neg A \sqcap \forall PO.\neg A, \\ C_{\text{joint}} = \exists PP.(\forall PPI.A \sqcap \forall PO.A) \sqcap \exists PO.\neg A.$$

Both are unsatisfiable but accepted by that one-sided test. The first excludes all four possible relations between the root and an A -point sharing a part; the second excludes both possibilities in $PPI \circ PO = \{PPI, PO\}$ between its two witnesses.

On the separate full-language certificate route, the current state is

$$\begin{aligned} \text{F6} &\xrightarrow{?} \text{refined uniformization} \\ &\xrightarrow{?} \text{complete candidate lists} \implies \text{decidability.} \end{aligned}$$

Only the last implication, with the required premise supplied, is the certified conditional decision result described by the current overview. The two new theorems do not discharge either question-marked implication.

The next useful target is a finite representation theorem, or a counterexample to a precisely defined representation scheme, that handles forced common-part overlaps together with required optional

overlaps. A pair-matrix extension must coordinate the relations among witnesses; checking its success on C_{bad} and C_{joint} is a necessary diagnostic, not a completeness proof. Conversely, disproving a particular width bound would not by itself establish undecidability; that would require a reduction from an undecidable problem.

Final assessment. The present $\forall\text{PO}$ -free argument survived this audit. The new results give a controlled order-only normalization and a further decidable syntactic slice. No proof of decidability or undecidability of unrestricted $\mathcal{ALCI}_{\text{RCC5}}$ was obtained.

8 Outlook: could more capable AI resolve the problem soon?

Judgment and horizon. Taking “near future” to mean approximately the next two to five years, my assessment is **moderately optimistic**, with substantial uncertainty. I think an AI-assisted resolution in that interval is plausible enough to justify a focused research effort. I would not treat it as a likely deliverable on a fixed schedule. This is a subjective assessment of the research situation, not a calibrated forecast of future model capabilities. The evidence supports optimism about further useful results more strongly than it supports a timetable for a full solution.

Reasons for optimism. This project already has an unusually useful starting point for future models: a precise open question, a certified structural normal form, a certified decidable fragment, explicit counterexamples to attractive repairs, and a map of the current proof obligations. The fragment result shows that a change in model construction can remove difficulties that repeated local repairs had not resolved. A better model may find another such change, recognize an applicable theorem in an adjacent field, or discover a reduction whose required structure differs from the familiar grid attempts. The short normalization and erasure arguments here also show that there are still tractable structural consequences to extract from the present framework; their modest scope should not be mistaken for evidence that the final step is close.

Models with better mathematical reasoning, more reliable long-context work, and better use of executable search and proof assistants could explore these alternatives more systematically. In particular, finite countermodel search can reject proposed local invariants cheaply, while a proof assistant can prevent an accepted formal argument from silently assuming its own completeness premise. A well-maintained record of failed lemmas would help future attempts spend their effort on different ideas.

Reasons for caution. The remaining task is a finite-to-global existence or representation problem, or an undecidability reduction. The project has not exposed a missing routine calculation whose completion merely requires a larger computation. A proposed finite state must still capture all relevant interactions of an unbounded model. More fluent explanations, larger bounded tests, and repeated agreement among similar models do not establish that property. Closely related models can share the same mistaken abstraction, so their agreement is weaker evidence than a checked proof or a concrete counterexample.

Nor does the number of historical failures reliably measure how far the problem is from a solution. A new invariant could shorten the argument dramatically; equally, the surviving obstruction might resist several further generations of models. The current evidence does not justify a confident preference between eventual decidability and undecidability. The decidable fragments give positive structure without resolving the expressive power added by the unrestricted interactions.

What would increase my confidence. I would update the forecast substantially after a model produces one of three concrete advances: a complete finite or automatic representation theorem for the unrestricted semantics; a provably terminating procedure whose soundness and completeness cover the full language; or a faithful reduction from a known undecidable problem. The decisive intermediate lemmas should be formalized early, with semantic adequacy and both directions of any reduction reviewed independently. A polished proposed proof that postpones its main completeness lemma would provide little reason to update the forecast.

My practical expectation is therefore that increasingly capable AI, combined with sustained human direction and formal checking, makes this a worthwhile target for continued work. I am hopeful about a resolution within a few years, but the present report supplies no mathematical basis for promising one. The part of the workflow most likely to help is a cycle of distinct structural hypotheses, adversarial counterexamples, and early certification of the lemmas that survive.

A Source map and finite checks

The following references are to the supplied snapshot, not to an assumed future state of the public repository. Line numbers are approximate navigation aids; declaration names are the more stable identifiers.

Current artifact or declaration	Location in the supplied source
Ordered-disjoint normal form	Fragment paper, lines 318–443.
Cone definitions and correctness	Fragment paper, lines 444–693 and 814–917.
Full-language open premises	Overview, lines 2373–2418.
<code>coneScheme_complete</code>	<code>P0FreeLift.lean</code> , near line 42317.
<code>unf_truth</code>	Same file, near lines 44006–44050.
<code>coneScheme_sound</code>	Same file, near line 44093.
<code>decidableSat_cone</code>	Same file, near line 44196.
<code>decidableFSat</code>	Same file, near line 44589.
<code>satisfiable_iff_set</code>	Same file, near line 44948.
<code>decidableSetSat</code>	Same file, near line 44976.

Two finite checks supported the review. First, enumerating all triples of nonempty subsets of a four-element universe gave exactly the 25 composition-table cells printed in the current fragment paper. Second, enumerating the 41 ordered-disjoint frames on three labelled points and all eight valuations of A tested the boundary formulas in Section 6. Counting satisfying combinations of frame, valuation and designated root gave:

Boundary concept	Erased formula	Original formula
$C_{\exists PO}$	480	0
$C_{\exists PP}$	12	0
$C_{\forall DR}$	12	0

These counts are finite corroboration. The all-cardinality unsatisfiability claims follow from the explicit relation case splits, and the normalization and erasure theorems follow from their proofs, not from bounded search. No kernel certification of the new results is claimed.

References

- [1] *ALCI_RCC5 decidability project*. Supplied snapshot `alcircc5-master(1).zip`, reviewed 7 October 2026. Current status sources: `README.md`, `approaches/ConeScheme/Overview.md`, and `approaches/Lean_F6/Overview.md`. Public repository: <https://github.com/lambda-mikel/alcircc5>.
- [2] *Current focused paper on the $\forall$ PO-free fragment*. In the supplied project snapshot: `papers/pofree_fragment_arxiv.tex` and its PDF.
- [3] *Current project overview paper*. In the supplied project snapshot: `papers/overview_arxiv.tex` and its PDF.
- [4] *Current Lean development*. In the supplied project snapshot: `formal/POFreeLift.lean`. Full-language conditional pipeline: `formal/Round19Transport.lean`.
- [5] C. Lutz and F. Wolter. Modal logics of topological relations. *Logical Methods in Computer Science*, 2(2:5):1–41, 2006. [https://doi.org/10.2168/LMCS-2\(2:5\)2006](https://doi.org/10.2168/LMCS-2(2:5)2006).
- [6] M. Wessel. *Qualitative spatial reasoning with the ALCI_RCC family: first results and unanswered questions*. Technical Report FBI-HH-M-324/03, Universität Hamburg, 2002/2003. Included in the supplied snapshot as `papers/report7.pdf`.